

Amazon RDS – Primary & Read Replica Architecture

Hello Everyone, Subham here!

Today I explored an advanced concept in Amazon RDS – Primary & Read Replica architecture.

◆ Primary DB Instance

- Supports both Read and Write operations.
- All new data (INSERT, UPDATE, DELETE) happens here.
- Also used for critical reads (like Order Confirmation) where accuracy matters.

◆ Read Replica

- Data is asynchronously replicated from the Primary.
- Supports only Read queries.
- Ideal for scaling read-heavy traffic (product browsing, reports, analytics).

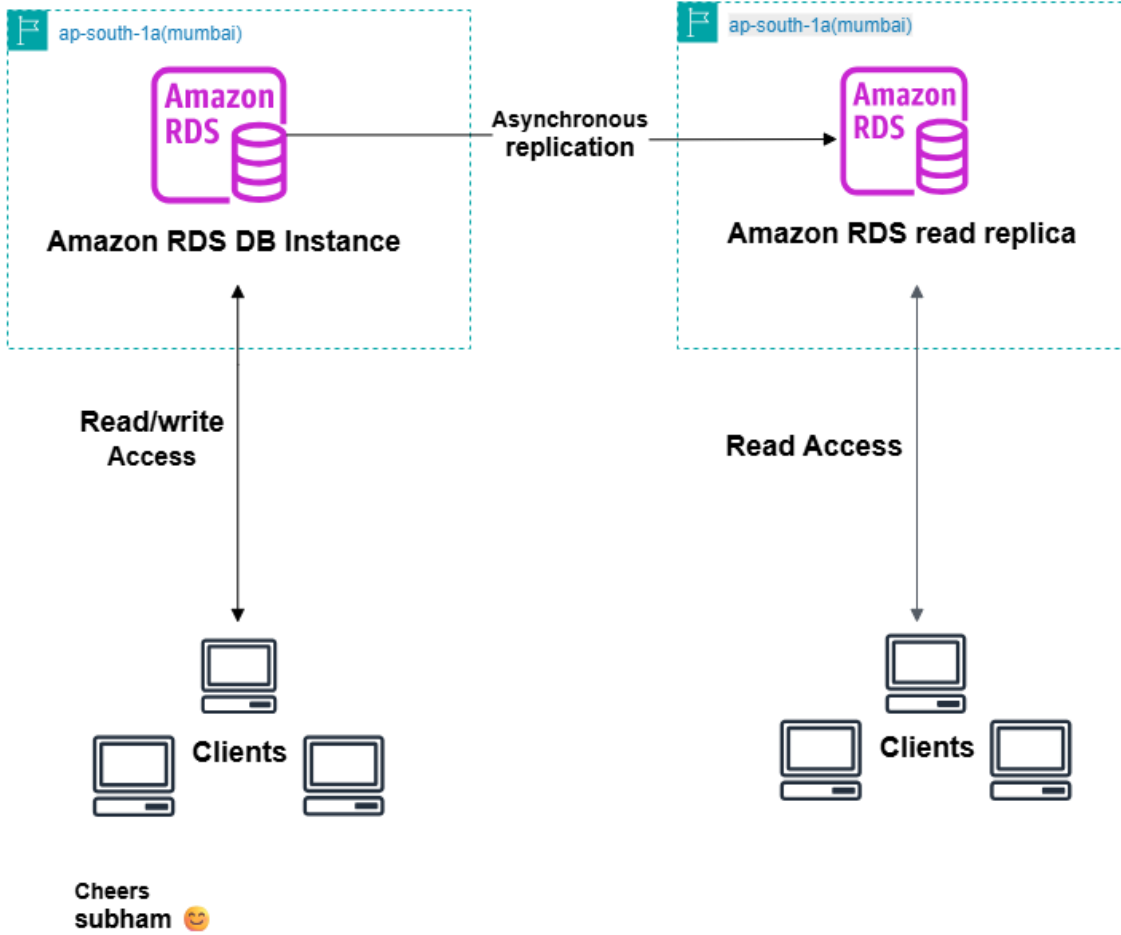
📌 Flipkart Example:

- When a user searches for a mobile → Query goes to Read Replica (fast, scalable).
- When a user places an order → Write + Read both happen in Primary DB (write the order, then read back for confirmation).
- When a user later checks order history → That read can come from a Replica

🌟 This combination ensures performance, scalability, and high availability. If the Primary fails, a Replica can even be promoted to a new Primary.

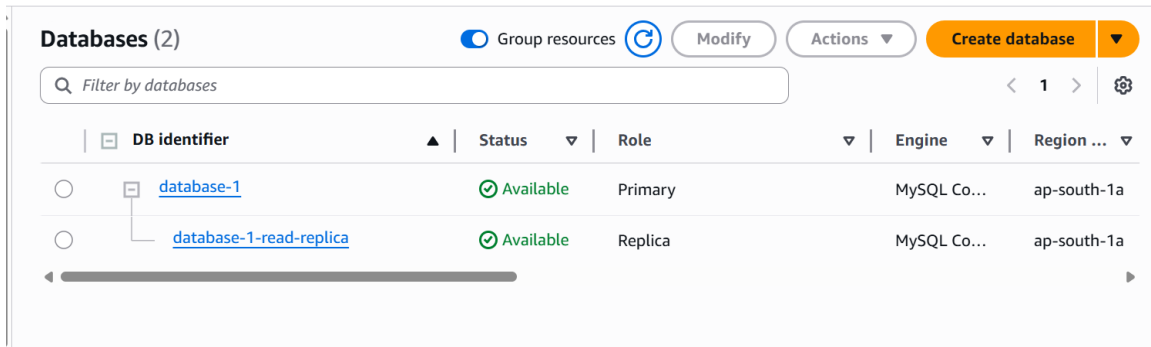


AWS Cloud



Step-1 : Create aws rds data base

. after available the database then select the data base and create read replica of that database .



◆ Step 2: Launch EC2 Instance

1. Go to **AWS Console** → **EC2** → **Launch Instance**.
2. Choose **Amazon Linux 2023 AMI** (or Ubuntu).
3. Select instance type (t2.micro for free tier).
4. Configure **Security Group**:
 - Allow **SSH (22)** from your IP.
 - Allow **MySQL/Aurora (3306)** from your IP **and** RDS security group.

Launch and connect to your EC2 using instance direct connect

◆ Step 3: Install MySQL Client on EC2

For **Amazon Linux 2023**:

```
sudo dnf install mariadb105 -y
```

Check version:

```
mysql --versio
```

◆ Step 4: Connect to Primary RDS DB

1. From RDS console, copy the **Primary Endpoint**.
2. Connect from EC2:
3. `mysql -h <primary-endpoint> -u <username> -p`
4. Create a database & table:
5. `CREATE DATABASE subham;`
6. `USE subham;`

```
MySQL [ (none) ]> show databases;
+-----+
| Database                |
+-----+
| information_schema      |
| mysql                   |
| performance_schema      |
| shell                   |
| subha                   |
| subham                  |
| sys                     |
| test                    |
+-----+
8 rows in set (0.001 sec)

MySQL [ (none) ]> 
```

7. `CREATE TABLE student (id INT, name VARCHAR(50));`
 8. `INSERT INTO student VALUES (1, 'Ravi');`
-

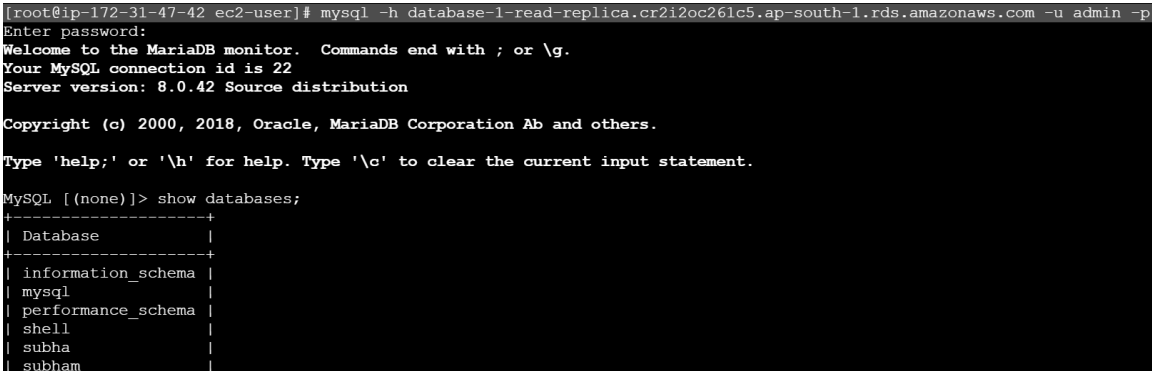
◆ Step 5: Verify Data in Primary

SHOW DATABASES;

SELECT * FROM student;

◆ Step 6: Connect to Read Replica

1. From RDS console, copy the **Replica Endpoint**.
2. Connect from EC2:
3. `mysql -h <replica-endpoint> -u <username> -p`
4. Check replication:
5. SHOW DATABASES;
6. USE subham;
7. SELECT * FROM student;

8. A terminal window showing a MySQL connection from an EC2 instance to an Amazon RDS Read Replica. The command used is `mysql -h database-1-read-replica.cr2i2oc26lc5.ap-south-1.rds.amazonaws.com -u admin -p`. The terminal output shows the MySQL welcome message, the connection ID (22), and the server version (8.0.42). The user then runs `show databases;` and the output lists the following databases: information_schema, mysql, performance_schema, shell, subha, and subham.

```
[root@ip-172-31-47-42 ec2-user]# mysql -h database-1-read-replica.cr2i2oc26lc5.ap-south-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 22
Server version: 8.0.42 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| shell |
| subha |
| subham |
+-----+
```

✓ You will see the data inserted in **Primary**.

◆ Step 7: Test Read-Only Restriction

```
Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| subha |
| subham |
| sys |
| test |
+-----+
7 rows in set (0.002 sec)

MySQL [(none)]> create database kal;
ERROR 1290 (HY000): The MySQL server is running with the --read-only option so it cannot execute this statement
MySQL [(none)]>
```

i-0186f24f7e64d00f4 (test-db) ✕

PublicIPs: 3.111.144.44 PrivateIPs: 172.31.47.42

Try creating DB on Replica:

CREATE DATABASE test2;

👉 You will get an error (read-only mode).

✅ Final Behavior

- **Primary DB** → Read + Write.
- **Replica DB** → Only Read (no Create/Insert/Update/Delete).

Thank you

Stay connect

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